MAXIMAL OPEN OF THE BASE OVER WHICH THE ACTION MORPHISM IS UNRAMIFIED OR LOCALLY QUASI-FINITE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 05W2: sources/stacks-chow/tags.txt, source lines 7571–7571. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0ABM: sources/stacks-chow/tags.txt, source lines 13025–13025. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DSI: sources/stacks-chow/tags.txt, source lines 17258–17258. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let G be a group algebraic space over B. Assume $G \to B$ is locally of finite type.*

1. *There exists a maximal open subspace $U \subset B$ such that $G_U \to U$ is unramified and formation of U commutes with base change.*
2. *There exists a maximal open subspace $U \subset B$ such that $G_U \to U$ is locally quasi-finite and formation of U commutes with base change.*

Proof. By Morphisms of Spaces, Lemma 05W2 (resp. Morphisms of Spaces, Lemma 0ABM) there is a maximal open subspace $W \subset G$ such that $W \to B$ is unramified (resp. locally quasi-finite). Moreover formation of W commutes with base change. By Lemma 0DSI we see that $U = e^{-1}(W)$ in either case. □

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